

Exercises 2A:

1) Find a list of four distinct vectors in F^3 whose span equals

$$U = \{(x, y, z) \in F^3 \mid x + y + z = 0\}$$

all three vectors need to be in U as the span includes

$$a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$$

where $a_k \neq 0$ for all others do.

So a simple vector could be $(1, -1, 0)$

$$1 + (-1) + 0 = 0$$

$$\text{So } v_1 = (1, -1, 0)$$

then we need to add another v_2

so that it is also in U

$$\text{Let } v_2 = (1, 0, -1)$$

now our summed vector is

$$\begin{aligned} a_1 v_1 + a_2 v_2 &= a_1(1, -1, 0) + a_2(1, 0, -1) \\ &= (a_1 + a_2, -a_1, -a_2) \end{aligned}$$

which is still in U

now

choosing $v_3 = (0, 1, -1)$

we get the vector

$$a_1 v_1 + a_2 v_2 + a_3 v_3 = (a_1 + a_2, a_3 - a_1, -a_2 - a_3)$$

which is also in U

now adding $v_4 = (0, 0, 0)$

which is also in U

$$\begin{aligned} \text{the sum } v &= a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 \\ &= (a_1 + a_2, a_3 - a_1, -a_2 - a_3) \end{aligned}$$

$$a_1 + a_2 + a_3 - a_1 - a_2 - a_3 = 0$$

$$\forall a_1, a_2, a_3, a_4 \in F$$

$$\therefore \text{Span}(v_1, v_2, v_3, v_4) \subseteq U$$

Proving the other way around.

$$\text{Let } x = a_1 + a_2$$

$$\text{and } y = a_3 - a_1$$

this is also
allowed as y
can still take
any value

by setting $a_3 = \text{value} + a_1$

(since this way x

can be any value

Since for all x

let $a_1 = \frac{x}{2}$ and $a_2 = \frac{x}{2}$)

$$\text{now } \forall x, y, z \in U \quad x + y + z = 0$$

$$\therefore z = -x - y = -a_1 - a_2 - a_3 + a_1 = -a_2 - a_3$$

$$\therefore z = -a_2 - a_3$$

$$\therefore U \subseteq \text{Span}(v_1, v_2, v_3, v_4)$$

$$\therefore U = \text{Span}(v_1, v_2, v_3, v_4)$$

2) Prove or give a Counter example: If v_1, v_2, v_3, v_4 spans V then the list:

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

also spans V .

$$V = \left\{ a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 \mid a_1, \dots, a_4 \in F \right\}$$

$$c_1 v_1 - c_1 v_2 + c_2 v_2 - c_2 v_3 + c_3 v_3 - c_3 v_4 + c_4 v_4$$

$$= c_1 v_1 + (c_2 - c_1) v_2 + (c_3 - c_2) v_3 + (c_4 - c_3) v_4$$

$$\text{let } c_1 = a_1, c_2 = a_2 + c_1, c_3 = a_3 + c_2, c_4 = a_4 + c_3$$

$$\therefore = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$$

$$\therefore V \subseteq \text{Span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$$

Since the span of the new list is a Subspace of V :

$$\therefore \text{Span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4) \subseteq V$$

$$\therefore \text{Span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4) = V$$

3) Suppose $v_1, \dots, v_n$ is a list of vectors in V .
 For $k \in \{1, \dots, n\}$, let

$$w_k = v_1 + \dots + v_k.$$

Show that $\text{Span}(v_1, \dots, v_n) = \text{Span}(w_1, \dots, w_n)$.

$$\text{Span}(v_1, \dots, v_n) = \{c_1 v_1 + \dots + c_n v_n, c_1, \dots, c_n \in F\}$$

$$\text{Span}(w_1, \dots, w_n) = \{a_1 w_1 + \dots + a_n w_n, a_1, \dots, a_n \in F\}$$

$$\begin{array}{l|l} \begin{array}{l} w_1 = v_1 \\ w_2 = v_1 + v_2 \\ w_3 = v_1 + v_2 + v_3 \\ \vdots \\ w_n = v_1 + \dots + v_n \end{array} & \text{So } \text{Span}(w_1, \dots, w_n) \\ & = \\ & (a_1 + \dots + a_n) v_1 \\ & + \\ & (a_2 + \dots + a_n) v_2 \\ & + \\ & \vdots \\ & (a_k + \dots + a_n) v_k \\ & + \\ & \vdots \\ & a_n v_n \end{array}$$

$$\begin{array}{l} \text{So let } a_1 = c_1 - (a_2 + \dots + a_n) \\ \text{and } a_2 = c_2 - (a_3 + \dots + a_n) \end{array}$$

$$\text{and } \forall a_k = c_k - (a_{k+1} + \dots + a_n)$$

$$\text{until } a_n = c_n$$

$$\therefore \text{Span}(v_1, \dots, v_n) \subseteq \text{Span}(w_1, \dots, w_n)$$

now let

$$c_1 = (a_1 + \dots + a_m)$$

$$\vdots$$
$$c_m = a_m$$

and

you get

$$\text{Span}(w_1, \dots, w_m) \subseteq \text{Span}(v_1, \dots, v_m)$$

$$\therefore \text{Span}(v_1, \dots, v_m) = \text{Span}(w_1, \dots, w_m)$$

(4) a) Show that a list of length one in a vector space is linearly independent $\iff$ the vector in the list is not 0.

Let v_i be the list of vectors and v_i be the vector

If $a_1 v_1 = 0$ and $v_1 \neq 0$ then

$a_1 = 0$ by definition of the additive identity in a vector space.

If $a_1 v_1 = 0$ and $a_1 \neq 0$ then $v_1 = 0$ by the same.

b) Show that a list of length two in a vector space is linearly independent $\iff$ neither of the two vectors in the list is a scalar mult of the other.

Let v_1, v_2 be the list of two vectors.
If $a_1 v_1 + a_2 v_2 = 0$

$$\text{then } a_1 v_1 = -a_2 v_2$$

If $a_1 \neq 0$ (def of linear independence)

$$\text{then } v_1 = \left(\frac{-a_2}{a_1} \right) v_2$$

$$\text{Since } a_2, a_1 \in F \quad \therefore \frac{-a_2}{a_1} \in F$$

$$\text{So } v_1 = c v_2, \quad c \in F$$

5) find a number f such that

$$(3, 1, 4), (2, -3, 5), (5, 9, f)$$

is not linearly independent in $\mathbb{R}^3$

$$(3, 1, 4) + b(2, -3, 5) + c(5, 9, f) = 0$$

$$\therefore 3 + 2b = -5c \quad \therefore 6c + 3 + \frac{2}{3} = -5c$$

$$1 - 3b = -9c \quad \therefore b = \frac{3c + \frac{1}{3}}{3}$$

$$4 + 5b = -fc \quad \therefore 11c = -\frac{11}{3}$$

$$4 - \frac{10}{3} = \frac{1}{3}f$$

$$c = -\frac{1}{3}$$

$$\therefore b = -\frac{2}{3}$$

$$\therefore f = 2$$

6) Show that the list $(2, 3, 1), (1, -1, 2), (7, 3, c)$ is linearly dependent in $F^3 \iff c=8$.

$$\text{If } (2, 3, 1) + a(1, -1, 2) + b(7, 3, c) = 0$$

$$\text{then } 2 + a + 7b = 0 \quad \therefore a = -7b - 2$$

$$\text{and } 3 - a + 3b = 0 \quad \therefore 3 + 7b + 2 + 3b = 0$$

$$\text{and } 1 + 2a + bc = 0 \quad 10b = -5$$

$$\therefore 1 + 3 - \frac{1}{2}c = 0$$

$$b = -\frac{1}{2}$$

$$\therefore a = \frac{7}{2} - 2$$

$$\frac{1}{2}c = 4 \quad c = 8$$

$$a = \frac{3}{2}$$

7) a) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{R}$, then the list $1+i, 1-i$ is linearly independent.

Let $a \in \mathbb{R}$

$$\text{If } a(1+i) = 1-i$$

then $a = i \notin \mathbb{R}$ by contradiction

$1+i, 1-i$ is not linearly independent.

b) Show that if we think of $\mathbb{C}$ as a vector space over $\mathbb{C}$ then the list $1+i, 1-i$ is linearly dependent.

$$i(1+i) = 1-i, \quad i \in \mathbb{C} \quad \therefore \text{linearly dependent.}$$

8) Suppose v_1, v_2, v_3, v_4 is linearly independent in V .
prove that the list

$$v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$$

is also linearly independent.

$$\text{Let } v_1, v_2, v_3, v_4 \in V$$

$$\text{Such that } v_1, a_1, \dots, a_4 \in F \neq 0$$

$$\text{and } a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = b \neq 0$$

$$\text{If } c_1, c_2, c_3, c_4 \in F$$

then let $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ be linearly dependent

Such that:

$$c_1(v_1 - v_2) + c_2(v_2 - v_3) + c_3(v_3 - v_4) + c_4 v_4 = 0$$

$$\therefore c_1 v_1 + (c_2 - c_1) v_2 + (c_3 - c_2) v_3 + (c_4 - c_3) v_4 = 0$$

now let $C_1 = a_1, C_2 = a_2 + C_1, C_3 = a_3 + C_2, C_4 = a_4 + C_3$

$$\therefore = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4 = 0$$

which is a contradiction, $\therefore$ the list is linearly independent.

9) Prove or give a counter example: If $v_1, v_2, \dots, v_m$ is linearly independent list of vectors in V , then

$5v_1 - 4v_2, v_2, v_3, \dots, v_m$
is linearly independent.

if $a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$ then $a_1, \dots, a_m = 0$

$$\text{Let } v_1 = \left(\frac{4}{5}, 1\right), v_2 = \left(1, \frac{5}{4}\right), v_k = (1, 1)$$

$$\text{Therefore } a_1 v_1 + \dots + a_m v_m \neq 0$$

$$\text{however } 5v_1 - 4v_2 = (4, 5) - (4, 5)$$

$$\therefore a_1 (5v_1 - 4v_2) + a_2 v_2 + \dots + a_m v_m = 0$$

$$\text{When } a_2 = 0$$

$$\text{and } a_1 \text{ and } a_3, \dots, a_m \neq 0$$

So it is not linearly independent.

(10) Prove or give a counterexample: If

$v_1, v_2, v_3, v_4, \dots, v_m$ is a linearly independent list of vectors in V and $\lambda \in F$ with $\lambda \neq 0$ then

$\lambda v_1, \lambda v_2, \dots, \lambda v_m$ is linearly independent.

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0 \quad \therefore a_1, a_2, \dots, a_m = 0$$

by def of linear independence

with $a_1, a_2, \dots, a_m \in F$

Now If $a_1 \lambda v_1 + a_2 \lambda v_2 + a_3 \lambda v_3 + \dots + a_m \lambda v_m = 0$

Where $a_i \neq 0$ and $\lambda \neq 0$

with $a_1, a_2, \dots, a_m, \lambda \in F$

then letting $a_1 \lambda = c_1, a_2 \lambda = c_2, \dots, a_m \lambda = c_m$

then with $c_1, c_2, c_3, \dots, c_m \in F$

$$c_1 v_1 + c_2 v_2 + \dots + c_m v_m = 0$$

Since $\lambda \neq 0$ $a_k \neq 0 \quad \therefore c_k \neq 0$ which is a contradiction.

$\therefore \lambda v_1, \lambda v_2, \lambda v_3, \dots, \lambda v_m$ is linearly independent

11) Prove or give a counterexample: If $v_1, \dots, v_m$ and $w_1, \dots, w_m$ are linearly independent lists of vectors in V , then the list:

$v_1 + w_1, \dots, v_m + w_m$ is linearly independent!

Let $v_1 = (1, 0)$, $v_2 = (0, 1)$ } from where
and $w_1 = (0, 1)$, $w_2 = (1, 0)$ } solutions

$$\therefore v_1 + w_1 = (1, 1), v_2 + w_2 = (1, 1)$$

$$\therefore v_1 + w_1, v_2 + w_2 = (1, 1), (1, 1)$$

$$a_1(1, 1) + a_2(1, 1) = 0$$

$$\text{When } a_1 = 1, a_2 = -1$$

So not linearly independent.

12) Suppose $v_1, \dots, v_m$ is linearly independent in V and that $w \in V$

Prove that if $v_1 + w, \dots, v_m + w$ is linearly dependent then $w \in \text{Span}(v_1, \dots, v_m)$

if

$$a_1 v_1 + \dots + a_m v_m = 0 \quad \text{then } a_1, a_2, \dots, a_m = 0$$

$$a_1, a_2, \dots, a_m \in F$$

$$\text{Span}(v_1, \dots, v_n) = a_1 v_1 + a_2 v_2 + \dots + a_n v_n$$

for $c_1, c_2, \dots, c_n \in F$

$$c_1(v_1 + \omega) + c_2(v_2 + \omega) + \dots + c_n(v_n + \omega) = 0$$

$$= c_1 v_1 + c_2 v_2 + \dots + c_n v_n + (c_1 + c_2 + \dots + c_n) \omega = 0$$

$$- (c_1 + c_2 + \dots + c_n) \omega = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$$

If $c_1, c_2, \dots, c_n \neq 0$ then RHS $\neq 0$

$$\therefore - (c_1 + c_2 + \dots + c_n) \neq 0 \text{ and } \omega \neq 0$$

$$\therefore \omega = \left(\frac{c_1}{-(c_1 + c_2 + \dots + c_n)} \right) v_1 + \left(\frac{c_2}{-(c_1 + c_2 + \dots + c_n)} \right) v_2 + \dots + \left(\frac{c_n}{-(c_1 + c_2 + \dots + c_n)} \right) v_n$$

$$\frac{c_k}{-(c_1 + c_2 + \dots + c_n)} \in F$$

$$\text{Let } a_1 = \frac{c_1}{-(c_1 + c_2 + \dots + c_n)}, a_2 = \frac{c_2}{-(c_1 + c_2 + \dots + c_n)}, \dots, a_n = \frac{c_n}{-(c_1 + c_2 + \dots + c_n)}$$

Since $a_1 v_1 + a_2 v_2 + \dots + a_n v_n \in \text{Span}(v_1, v_2, \dots, v_n)$

$$\therefore \omega \in \text{Span}(v_1, v_2, \dots, v_n)$$

13) Suppose $v_1, \dots, v_n$ is linearly independent in V and $\omega \in V$. Show that $v_1, \dots, v_n, \omega$ is linearly independent $\iff \omega \notin \text{Span}(v_1, \dots, v_n)$

If $a_1 v_1 + \dots + a_n v_n = 0$ for $(a_1, a_2, \dots, a_n) \in F$
then $a_1, a_2, \dots, a_n = 0$ (1)

If $v_1, \dots, v_n, \omega$ is linearly independent

and If $\omega \in \text{Span}(v_1, \dots, v_n)$

then $\omega = \lambda_1 v_1 + \dots + \lambda_n v_n$

$$a_1 v_1 + \dots + a_n v_n + \lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

$$= (a_1 + \lambda_1) v_1 + \dots + (a_n + \lambda_n) v_n = 0$$

$$\therefore a_1 + \lambda_1, a_2 + \lambda_2, \dots, a_n + \lambda_n = 0$$

by def of linear independence.

$$\text{If } a_1, \dots, a_n = 0$$

$$\text{then } \lambda_1, \dots, \lambda_n = 0$$

$$\text{and } \omega = 0$$

however if $\omega = 0$

$$\text{then } c_1 v_1 + \dots + c_n v_n + c_{n+1} \omega = 0$$

Could have any value of c_{n+1}
and the list $\cup$ could not be
linearly independent.

If $a_1, \dots, a_n \neq 0$

$$\text{then } \lambda_1 = -a_1, \lambda_2 = -a_2, \dots, \lambda_n = -a_n$$

but then any multiple of both a_k and
 λ_k works $\cup$ for $k \in \{1, 2, \dots, n\}$

$$(a_1 + \lambda_1) v_1 + \dots + (a_n + \lambda_n) v_n = 0$$

and so the list is not linearly independent

$\therefore$ by contradiction If $v_1, \dots, v_n, \omega$
is independent and given $v_1, \dots, v_n$ is independent
then $\omega \notin \text{span}(v_1, \dots, v_n)$

14) Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let $w_k = v_1 + \dots + v_k$. Show that the list $v_1, \dots, v_m$ is linearly independent $\iff$ the list $w_1, \dots, w_m$ is linearly independent.

If $v_1, \dots, v_m$ is linearly independent

such that for $a_1, \dots, a_m \in F$

$$a_1 v_1 + \dots + a_m v_m = 0$$

$$\text{then } a_1, \dots, a_m = 0 \quad (1)$$

the list $w_1, \dots, w_m = v_1, v_1 + v_2, \dots, v_1 + \dots + v_m$

for $c_1, \dots, c_m \in F$

$$\text{If } c_1 v_1 + \dots + c_m (v_1 + \dots + v_m) = 0$$

$$\therefore = (c_1 + \dots + c_m) v_1 + (c_2 + \dots + c_m) v_2 + \dots + c_m v_m$$

Since $c_1 + \dots + c_m, \dots, c_m \in F$

then $(c_1 + \dots + c_m), \dots, c_m = 0$ by (1)

$\therefore w_1, \dots, w_m$ is linearly independent

15) Explain why there does not exist a list of six polynomials that is linearly independent in $\mathcal{P}_4(F)$.

Length of linearly independent list
 $\leq$
Length of spanning list

If $n > 0$ then $\mathcal{P}_n(F) = \text{Span}(1, z, \dots, z^n)$

So for $n=4$

$$\text{Span}(1, z, z^2, z^3, z^4) = a_0 1 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4$$

$$a_0, \dots, a_4 \in F$$

$$\forall z \in F$$

Since the spanning list is 5 long then any list over S cannot be linearly independent.

16) Explain why no list of 4 polynomials spans $\mathcal{P}_4(F)$.

Let the list $1, z, z^2, z^3$ be a list of length 4 polynomials.

$$\text{the } \text{Span}(1, z, z^2, z^3) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$$
$$a_0, \dots, a_3 \in F \text{ and } \forall z \in F$$

$$\mathcal{P}_4(F) = \text{Span}(1, \dots, z^4) > \text{Span}(1, \dots, z^3)$$

??, $\Rightarrow$
more needed!

17) Prove that V is infinite-dimensional $\Leftrightarrow$ there is a sequence $v_1, v_2, \dots$ of vectors in V such that $v_1, \dots, v_n$ is linearly independent for every positive integer n .

A vector space V is called infinite-dimensional if no list of vectors in V spans the space.

If there is no list $v_1, \dots, v_{n+1} \in V$ such that $v_1, \dots, v_n$ is linearly independent, but $\forall v_{n+1} \in V$

$$c_1 v_1 + \dots + c_n v_n = v_{n+1} \quad c_1, \dots, c_n, c \in F$$

$$c_1, \dots, c_n, c \neq 0$$

$$\therefore v_{n+1} = c_1 v_1 + \dots + c_n v_n \quad c_1, \dots, c_n \in F$$

$$c_1, \dots, c_n \neq 0$$

then $a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1}$

$$= (a_1 + c_1 a_{n+1}) v_1 + \dots + (a_n + c_n a_{n+1}) v_n$$

$$\therefore \text{Contained in } \text{Span}(v_1, \dots, v_n)$$

and therefore $v_1, \dots, v_n$ spans V and V is not infinite dimensional.

If $v_1, \dots, v_{n+1} \in V$ is linearly independent
 $\forall n \in \mathbb{Z}^+$ then $v_{n+1} \notin \text{Span}(v_1, \dots, v_n)$
and therefore $\forall v_1, \dots, v_n \text{ Span}(v_1, \dots, v_n) \neq V$
and V is infinite dimensional.

Feels a little weak maybe go back
over / check.

18) Prove that F^∞ is infinite-dimensional.

Let there be a list $x_1, x_2, \dots, x_n \in F$

Such that $x_1, x_2, \dots, x_n$ spans F^∞

$$\text{i.e. } \left\{ a_1 x_1 + a_2 x_2 + a_3 x_3 + \dots + a_n x_n \mid \forall a_1, \dots, a_n \in F \right\} = F^\infty$$

then let $x_{n+1} \in F$, $x_{n+1} \notin x_1, x_2, x_3, \dots, x_n$

So that $x_1, x_2, \dots, x_{n+1}$ is linearly independent in F^∞

Since $x_{n+1} \in F$ then $x_{n+1} \in F^\infty$

and $x_1, x_2, x_3, \dots, x_n$ does not span F^∞

this can also be done for $x_1, \dots, x_{n+1}$ and on forever as there will always be linearly independent scalars in F^∞

19) Prove that the real vector space of all continuous real valued functions on the interval $[0,1]$ is infinite dimensional.

$$\text{Let } U = \{f \mid f: \mathbb{R}^{[0,1]} \rightarrow \mathbb{R}\}$$

then for $f_1, f_2, \dots, f_n \in U$

$$\text{Span}(f_1, f_2, \dots, f_n) = a_1 f_1(x) + a_2 f_2(x) + \dots + a_n f_n(x)$$

$$\forall x \in [0,1], \forall a_1, a_2, \dots, a_n \in \mathbb{R}$$

Since there exists $g \in U$ such that

$$g(x) \neq (a_1 f_1(x) + \dots + a_n f_n(x)) \quad \forall x \in [0,1], a_1, \dots, a_n, c \in \mathbb{R}$$

then $\text{Span}(f_1, f_2, \dots, f_n) \neq U$

20) Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(F)$ such that $p_k(z) = 0$ for each $k \in \{0, \dots, m\}$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(F)$.

Let $P = p_0, p_1, \dots, p_m \in \mathcal{P}_m(F)$

such that $p_k(z) = 0, \forall k \in \{0, \dots, m\}$

for $a_0, a_1, \dots, a_m \in F, \forall z \in F$

Let $a_0 p_0(z) + a_1 p_1(z) + \dots + a_m p_m(z) = 0 \Leftrightarrow a_0, a_1, \dots, a_m = 0$

for $z = z$

$$a_0 p_0(z), a_1 p_1(z), \dots, a_m p_m(z) = 0$$

$\forall a_0, a_1, \dots, a_m \in F$

So it is not true that P is linearly independent in $\mathcal{P}_m(F)$